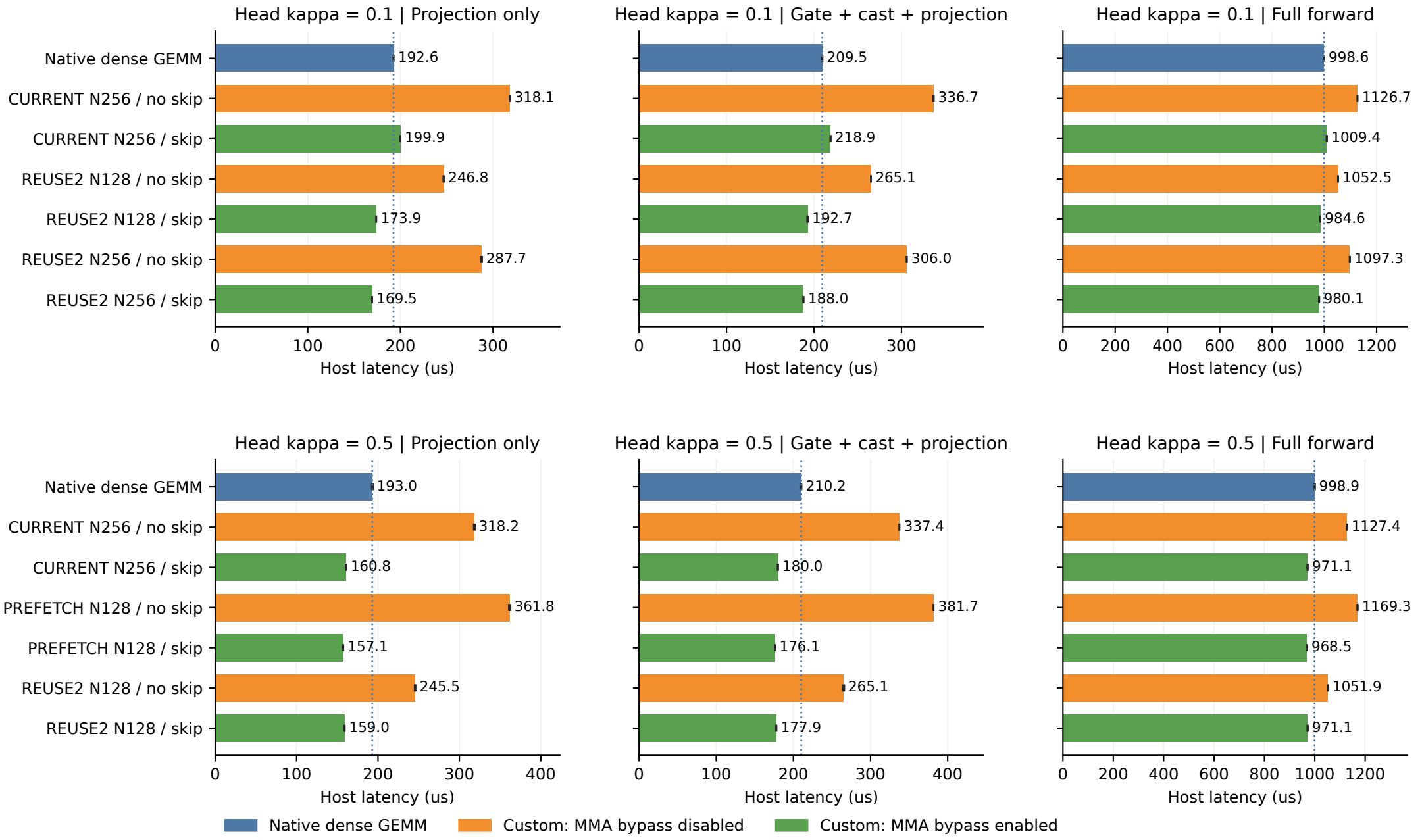
Qualified head implementations: the dense efficiency gap shrinks, full gains remain modest
14M | B1/T2048 | all logits | same gated checkpoints | 3 fresh processes on one RTX 5090
Head kappa = 0.1 | Projection only
Native dense GEMM
CURRENT N256 / no skip
CURRENT N256 / skip
REUSE2 N128 / no skip
REUSE2 N128 / skip
REUSE2 N256 / no skip
REUSE2 N256 / skip
192.6
318.1
199.9
246.8
173.9
287.7
169.5
0
100
200
300
Host latency (us)
Head kappa = 0.1 | Gate + cast + projection
209.5
336.7
218.9
265.1
192.7
306.0
188.0
0
100
200
300
Host latency (us)
Head kappa = 0.1 | Full forward
998.6
1126.7
1009.4
1052.5
984.6
1097.3
980.1
0
200
400
600
800
1000
1200
Host latency (us)
Head kappa = 0.5 | Projection only
Native dense GEMM
CURRENT N256 / no skip
CURRENT N256 / skip
PREFETCH N128 / no skip
PREFETCH N128 / skip
REUSE2 N128 / no skip
REUSE2 N128 / skip
193.0
318.2
160.8
361.8
157.1
245.5
159.0
0
100
200
300
400
Host latency (us)
Head kappa = 0.5 | Gate + cast + projection
210.2
337.4
180.0
381.7
176.1
265.1
177.9
0
100
200
300
400
Host latency (us)
Head kappa = 0.5 | Full forward
998.9
1127.4
971.1
1169.3
968.5
1051.9
971.1
0
200
400
600
800
1000
1200
Host latency (us)
Native dense GEMM
Custom: MMA bypass disabled
Custom: MMA bypass enabled
Bars: process geometric means. Whiskers: process min/max, not confidence intervals. Boundaries overlap; do not sum.